

Mathematics A

Unit 2 • Chapter OL-T53 Classified

Cross-session • chapter-organised candidate

39 classified items • 117 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T53 • 39 items • 117 marks

Chapter	Items	Marks	Starts
01. OL-T53 Statistics Toolkit	39	117	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Statistics Toolkit

Unit 2 • 117 marks • 39 item(s)

June 2025 | 4MA1-1H Q1 | 5 marks

Estimate the mean from grouped intervals

June 2025 | 4MA1-2H Q1 | 3 marks

Construct a data set from stated averages and range

June 2024 | 4MA1-2H Q1 | 3 marks

Construct a data set from stated averages and range

June 2024 | 4MA1-2H Q12 | 3 marks

Combine or separate subgroup means

November 2023 | 4MA1-2H Q1 | 1 marks

Identify the modal class

November 2023 | 4MA1-2H Q1 | 3 marks

Estimate a grouped total rather than the mean

November 2023 | 4MA1-2H Q8 | 3 marks

Find a required score for a target mean

November 2023 | 4MA1-2H Q13 | 2 marks

Find quartiles and IQR from discrete data

June 2023 | 4MA1-1H Q2 | 3 marks

Estimate a grouped total rather than the mean

June 2023 | 4MA1-1H Q10 | 2 marks

Find quartiles and IQR from discrete data

June 2023 | 4MA1-2H Q10 | 4 marks

Combine or separate subgroup means

November 2024 | 4MA1-1H Q1 | 1 marks

Identify the modal class

November 2024 | 4MA1-1H Q1 | 4 marks

Estimate the mean from grouped intervals

November 2024 | 4MA1-2H Q8 | 3 marks

Recover a missing value or frequency from a mean

November 2024 | 4MA1-2H Q10 | 5 marks

Read values and compare categories from a basic diagram

... 24 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

- 1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	Frequency
$0 < d \leq 5$	26
$5 < d \leq 10$	40
$10 < d \leq 15$	16
$15 < d \leq 20$	10
$20 < d \leq 25$	8

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean distance.

..... km
(4)

WORKING SPACE

- 1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

WORKING SPACE

Question 12

4MA1-2H Q1

MODULAR UNIT 2

3 marks

- 1 Here are eight numbers written in order of size

$$h \quad 6 \quad 7 \quad 8 \quad j \quad 16 \quad k \quad k$$

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h , the value of j and the value of k

 $h = \dots\dots\dots$ $j = \dots\dots\dots$ $k = \dots\dots\dots$

WORKING SPACE

Question 17

4MA1-2H Q12

MODULAR UNIT 2

3 marks

12 Ali uses a fitness tracker to count the number of steps he walks each day for 7 days.

For the first 4 days, his mean number of steps is 11 800

For the next 3 days, his mean number of steps is 13 207

Work out his mean number of steps for the 7 days.

WORKING SPACE

Question 21

4MA1-2H Q1 (a)

MODULAR UNIT 2

1 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (a) Write down the modal class.

.....
(1)

WORKING SPACE

Working space for Question 21

Continue your method

UNIT 2

1 marks

WORKING SPACE

Question 22

4MA1-2H Q1 (b)

MODULAR UNIT 2

3 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (b) Work out an estimate for the total length, in minutes, of these telephone calls.

..... minutes
(3)

WORKING SPACE

Working space for Question 22

Continue your method

UNIT 2

3 marks

WORKING SPACE

Question 27

4MA1-2H Q8 whole

MODULAR UNIT 2

3 marks

- 8 The mean number of goals scored by a hockey team in 8 matches is 6
The team plays 2 more matches and scores k goals in each match.
The mean number of goals scored by the hockey team in the 10 matches is 7

Work out the value of k

$k = \dots\dots\dots$

WORKING SPACE

Question 28

4MA1-2H Q13 whole

MODULAR UNIT 2

2 marks

13 Robert asked 11 people how many meetings they attended last week.

Here are the results in numerical order.

1 2 4 6 6 8 11 12 13 14 17

Find the interquartile range of the number of meetings.

.....

WORKING SPACE

Question 02

4MA1-1H Q2

MODULAR UNIT 2

3 marks

- 2 The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	Frequency
$0 < M \leq 30$	5
$30 < M \leq 60$	6
$60 < M \leq 90$	8
$90 < M \leq 120$	9
$120 < M \leq 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

..... minutes

WORKING SPACE

Question 10

4MA1-1H Q10

MODULAR UNIT 2

2 marks

10 Here are the test marks of 15 students.

7 10 14 15 16 17 18 19 20 20 23 25 30 39 40

Find the interquartile range of these marks.

WORKING SPACE

Question 23

4MA1-2H Q10

MODULAR UNIT 2

4 marks

10 Team **A** and Team **B** take part in a quiz league.

After 11 rounds, Team **A** has a mean score per round of 17

After 9 rounds, Team **B** has a mean score per round of 18

Both teams take part in a further round.

After this round, both teams have a mean score per round of 18.5

In the further round, Team **A** scored more points than Team **B**.

How many more?

WORKING SPACE

Question 01

4MA1-1H Q1 M01

MODULAR UNIT 2

1 marks

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (a) Write down the modal class.

.....
(1)

WORKING SPACE

Question 02

4MA1-1H Q1 M02

MODULAR UNIT 2

4 marks

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

..... dollars
(4)

WORKING SPACE

Question 22

4MA1-2H Q8 M01

MODULAR UNIT 2

3 marks

- 8 Kim puts 4 bananas in a bag.
The mean weight of the 4 bananas in the bag is 145 grams.

Andy puts one more banana into the bag.
The mean weight of the 5 bananas in the bag is 142 grams.

Work out the weight of the banana that Andy puts into the bag.

..... grams

WORKING SPACE

Question 24

4MA1-2H Q10 M01

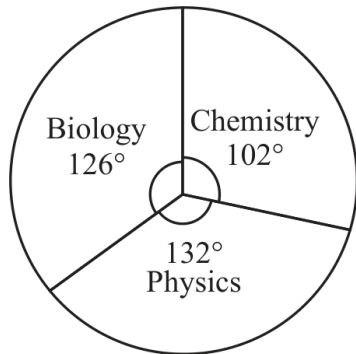
MODULAR UNIT 2

5 marks

- 10 All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.

The table shows information about the results for the year 11 students.

**Pie chart for year 10**

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10

There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

Question 24 (continued)

4MA1-2H Q10 M01

MODULAR UNIT 2

5 marks

WORKING SPACE

Question 20

4MA1-2H Q1 Q1

MODULAR UNIT 2

3 marks

- 1 Here are some integers where $a < b < c < d$

 $a \quad b \quad c \quad d \quad d \quad d$

The mode of the integers is 9

The median of the integers is 8

The range of the integers is 4

Work out the value of a , the value of b , the value of c and the value of d

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$ $d = \dots\dots\dots$

WORKING SPACE

Question 25

4MA1-2H Q12 Q12

MODULAR UNIT 2

3 marks

12 Larry is a delivery man.

He has 7 parcels to deliver.

The mean weight of the 7 parcels is 2.7 kg

Larry delivers 3 of the parcels.

Each of these 3 parcels has a weight of W kg

The mean weight of the other 4 parcels is 3.3 kg

Work out the value of W

 $W = \dots\dots\dots$

WORKING SPACE

Question 04

4MA1-1H Q5 Q5(a)

MODULAR UNIT 2

3 marks

5 Jenny has six cards.

Each card has a whole number written on it so that

the smallest number is 5

the largest number is 24

the median of the six numbers is 14

the mode of the six numbers is 8

Jenny arranges her cards so that the numbers are in order of size.

5					24
---	-------	-------	-------	-------	----

(a) For the remaining four cards, write on each dotted line a number that could be on the card.

(3)

Working space for Question 04

Continue your method

UNIT 2**3 marks**

WORKING SPACE

Question 05

4MA1-1H Q5 Q5(b)

MODULAR UNIT 2

3 marks

A basketball team plays 6 games.

After playing 5 games, the team has a mean score of 21 points per game.

After playing 6 games, the team has a mean score of 23 points per game.

(b) Work out the number of points the team scored in its 6th game.

(3)

WORKING SPACE

Question 23

4MA1-2H Q9

MODULAR UNIT 2

4 marks

- 9 The frequency table gives information about the number of points scored by a player.

Number of points	Frequency
0	13
1	17
2	8
3	x
4	11

The mean number of points scored is 2

Work out the value of x

$x =$

Working space for Question 23

Continue your method

UNIT 2
4 marks

WORKING SPACE

Question 28

4MA1-2H Q15

MODULAR UNIT 2

3 marks

- 15 Diyar recorded the distance, in kilometres, that he cycled each day for 11 days.
Here are his results.

8 10 12 13 5 23 21 7 5 16 14

Find the interquartile range of his results.

..... km

WORKING SPACE

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

- (a) Write down the modal class.

(1)

YOUR WORKING

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

(a) Write down the modal class.

(b) Work out an estimate for the mean weight of the 40 babies.

..... kg
(4)

YOUR WORKING

- 2 The table gives information about the speeds, in kilometres per hour, of 80 motorbikes as each pass under a bridge.

Speed (s kilometres per hour)	Frequency
$40 < s \leq 50$	10
$50 < s \leq 60$	16
$60 < s \leq 70$	19
$70 < s \leq 80$	23
$80 < s \leq 90$	12

- (a) Write down the modal class.

(1)

YOUR WORKING

- (b) Work out an estimate for the mean speed of the motorbikes as they pass under the bridge.
Give your answer correct to 3 significant figures.

..... kilometres per hour
(4)

YOUR WORKING

2 Ava writes down five whole numbers.

For these five numbers

the median is 7

the mode is 8

the range is 5

Find a possible value for each of the five numbers that Ava writes down.

YOUR WORKING

- 7 There are 5 cocoa pods in a bag.
The mean weight of the 5 cocoa pods is 398 grams.

A sixth cocoa pod is put into the bag.
The mean weight of the 6 cocoa pods is 401 grams.

Work out the weight of the sixth cocoa pod that is put into the bag.

..... grams

YOUR WORKING

12 15 people were asked how long, in minutes, they had been waiting for a bus.

Here are the results.

2 3 3 4 5 6 6 8 9 10 11 13 14 15 18

Find the interquartile range of these times.

..... minutes

YOUR WORKING

10 Jethro has sat 5 tests.

Each test was marked out of 100 and Jethro's mean mark for the 5 tests is 74

Jethro has to sit one more test that is also to be marked out of 100

Jethro wants his mean mark for all 6 tests to be at least 77

Work out the least mark that Jethro needs to get for the last test.

YOUR WORKING

14 Here is the number of goals that Henri's team scored one summer in each water polo match.

5 8 9 11 13 13 14 15 16 17 20

Find the interquartile range of the numbers of goals.
Show your working clearly.

YOUR WORKING

- 3 The table gives information about the amounts of money, in euros, that 70 of Anjali's friends spent last Saturday.

Money spent (S euros)	Frequency
$0 < S \leq 8$	6
$8 < S \leq 16$	14
$16 < S \leq 24$	19
$24 < S \leq 32$	25
$32 < S \leq 40$	6

One of Anjali's 70 friends is going to be chosen at random.

- (a) Find the probability that this friend spent more than 24 euros last Saturday.

.....
(1)

YOUR WORKING

- (b) Work out an estimate for the mean amount of money spent by Anjali's friends last Saturday.
Give your answer correct to 2 decimal places.

..... euros
(4)

YOUR WORKING

4 Here is a list of six numbers written in order of size.

4 7 x 10 y y

The numbers have

a median of 9

a mean of 11

Find the value of x and the value of y .

$x =$

$y =$

(Total for Question 4 is 4 marks)

YOUR WORKING

- 2 The table shows information about the lengths of time, in minutes, 120 customers spent in a supermarket.

Length of time (L minutes)	Frequency
$20 < L \leq 30$	6
$30 < L \leq 40$	26
$40 < L \leq 50$	31
$50 < L \leq 60$	40
$60 < L \leq 70$	17

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean length of time spent by the 120 customers in the supermarket.

.....minutes
(4)

(Total for Question 2 is 5 marks)

YOUR WORKING

Answer ALL TWENTY FOUR questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 80 students entered a dancing competition.

The table gives information about the length of time, in minutes, for which each student spent dancing.

Time (m)	Frequency
$0 < m \leq 12$	11
$12 < m \leq 24$	25
$24 < m \leq 36$	23
$36 < m \leq 48$	15
$48 < m \leq 60$	6

Work out an estimate for the mean length of time the students spent dancing.

..... minutes

(Total for Question 1 is 4 marks)

YOUR WORKING

- 15** Here is a list giving the numbers of runs scored last week by the eleven members of cricket team **A**.

2 3 4 6 21 26 27 32 34 61 72

The interquartile range of the numbers of runs scored last week by the eleven members of cricket team **B** was 42

Using a suitable calculation, write down one comparison between the numbers of runs scored by the members of cricket team **A** and the members of cricket team **B**.
Show your working clearly.

(Total for Question 15 is 3 marks)

YOUR WORKING

- 1 The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

Number of kilometres (K)	Frequency
$0 \leq K < 5$	8
$5 \leq K < 10$	7
$10 \leq K < 15$	3
$15 \leq K < 20$	10
$20 \leq K < 25$	2

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

..... kilometres

(Total for Question 1 is 4 marks)

YOUR WORKING

4 e, f and g represent integers such that

$e \quad f \quad 7 \quad g \quad 12 \quad 17$

is a list of integers written in order of size.

The integers have

- a range of 15
- a median of 8.5
- a mean of 9

Work out the value of e , the value of f and the value of g

$e = \dots\dots\dots$

$f = \dots\dots\dots$

$g = \dots\dots\dots$

(Total for Question 4 is 3 marks)

YOUR WORKING

16 Ivan asked 15 people how many books they read last year.

Here are his results.

1 1 3 4 6 7 8 9 10 12 15 25 30 37 50

Work out the interquartile range of the number of books read.

(Total for Question 16 is 2 marks)

YOUR WORKING
